MEDICAL DEVICE MARKETING
UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

1. Project Definition

Introduction

A **project** is a planned set of activities carried out to achieve a specific objective within a defined **scope, time, cost, and quality**.

In medical device development, a project may involve designing, developing, testing, validating, obtaining regulatory clearance/approval, and launching a new medical device.

Example

Development of an **IoT-based patient monitoring system** can be considered a project with:

- Objective – Develop a real-time patient monitoring system
- Duration – 12 months
- Budget – ₹20 lakh
- Team – Biomedical engineers, software engineers, clinicians and marketing personnel
- Deliverable – Tested and validated prototype/product

Major elements of project definition

1. **Project title**
2. **Problem statement**
3. **Objectives**
4. **Scope**
5. **Deliverables**
6. **Target customers**
7. **Timeline**
8. **Budget**
9. **Resources**
10. **Quality requirements**
11. **Regulatory requirements**
12. **Risk management**

Importance

A clear project definition helps the team understand **what has to be developed, why it is required, who will use it, how much it will cost, and when it should be completed**.

2. Financial Targets

Financial targets are the **specific financial goals established for a product or project**.

They help determine whether the product is commercially viable.

Important financial targets

1. Development budget

Maximum amount allocated for product development.

2. Manufacturing cost

Cost required to manufacture one unit.

3. Selling price

Price at which the product is sold.

4. Revenue target

Expected income from product sales.

5. Profit target

Expected profit from the product.

6. Break-even point

The sales level at which total revenue equals total cost.

7. Return on Investment (ROI)

Measures the return generated from an investment.

ROI = (Net Return / Investment) × 100

Example

Suppose:

- Manufacturing cost = ₹8,000/device
- Selling price = ₹15,000/device
- Expected sales = 1,000 devices

The company can estimate its revenue, contribution margin, operating costs, and expected profit before deciding whether to proceed with the project.

3. Physician Input

Introduction

Physician input refers to obtaining information, suggestions, feedback, and clinical requirements from doctors during medical product development.

Physicians are important stakeholders because they understand the **clinical problem, patient requirements, workflow, treatment procedures, and limitations of existing technologies**.

Importance of Physician Input

1. Identification of clinical problems

Physicians can identify problems that require technological solutions.

Example:
A physician may identify that existing patient monitoring systems provide information too late for early intervention.

2. Defining product requirements

Doctors can specify:

- Required parameters
- Accuracy
- Response time
- Clinical functionality
- Patient population
- Desired alerts

3. Product design

Physician feedback helps engineers develop a device suitable for actual clinical environments.

4. Usability

Doctors can evaluate:

- User interface
- Display
- Controls
- Workflow
- Ease of operation

5. Clinical validation

Physicians may participate in clinical evaluation and provide feedback regarding clinical performance.

6. Risk identification

Doctors can identify possible clinical hazards and inappropriate uses.

7. Product acceptance

A product developed with appropriate clinical input is more likely to be accepted by healthcare professionals.

Methods of Obtaining Physician Input

- Interviews
- Questionnaires
- Focus groups
- Clinical observations
- Product demonstrations
- Advisory panels
- Prototype evaluation
- Clinical studies

Example

For a **smart prosthetic limb**, engineers may develop the sensors and control system, while physicians and rehabilitation professionals can provide input regarding:

- Patient comfort
- Gait requirements
- Safety
- Socket fit
- Rehabilitation needs
- Clinical usability

4. FDA Approval

Introduction

The **U.S. Food and Drug Administration (FDA)** regulates medical devices marketed in the United States. FDA's Center for Devices and Radiological Health (CDRH) regulates firms that manufacture, repackage, relabel, or import medical devices for the U.S. market.

An important point for students is that **not every medical device receives an "FDA approval."** Depending on the device and regulatory pathway, the product may be **510(k) cleared, De Novo classified, PMA approved, or exempt from a premarket submission**.

Objectives of FDA Regulation

FDA regulation aims to ensure that medical devices:

- Are reasonably safe
- Are effective for their intended use

- Meet applicable regulatory requirements
- Are appropriately manufactured
- Have appropriate labeling
- Continue to be monitored after marketing

Step 1 – Identify the Intended Use

The first step is to clearly define:

- What the device does
- Who will use it
- Which patients will use it
- What disease/condition it addresses
- Whether it diagnoses, treats, prevents, monitors, or supports a condition

The intended use and indications for use are important because they influence the device's classification and regulatory pathway.

Step 2 – Determine Device Classification

FDA generally classifies medical devices into:

Class I – Low risk

Generally subject to **General Controls**.

Examples may include certain low-risk devices.

Class II – Moderate risk

Subject to **General Controls + Special Controls**.

Class III – High risk

Subject to **General Controls + Premarket Approval (PMA)** in applicable cases.

FDA states that classification is risk-based, with Class I generally representing the lowest risk and Class III the highest.

Class	Risk	Main Controls
Class I	Low	General Controls
Class II	Moderate	General + Special Controls
Class III	High	General Controls + PMA in applicable cases

Step 3 – Identify the Appropriate Regulatory Pathway

The major pathways students should know are:

A. 510(k) – Premarket Notification

The manufacturer generally demonstrates that the new device is **substantially equivalent** to a legally marketed predicate device.

The comparison generally considers:

- Intended use
- Technological characteristics
- Performance
- Safety

FDA explains that a 510(k) is a pathway based on substantial equivalence to an appropriate legally marketed device.

Important: The correct term is generally **"FDA clearance"**, not "FDA approval," for a 510(k).

B. De Novo Classification

The **De Novo pathway** can be used for certain novel devices for which there is no suitable predicate but whose risks can be appropriately controlled.

A successful De Novo request establishes a new classification for that device type. FDA identifies De Novo as a pathway for certain novel device types that have not previously been marketed in the United States.

C. Premarket Approval – PMA

PMA is the most stringent major premarket pathway.

It is generally associated with **Class III devices** requiring PMA.

The manufacturer must provide scientific evidence supporting the safety and effectiveness of the device. FDA describes PMA as the scientific and regulatory review used to evaluate the safety and effectiveness of applicable Class III devices.

Examples of devices that may require PMA include certain implantable and life-supporting devices.

D. HDE

Humanitarian Device Exemption (HDE) is associated with certain devices intended to benefit patients with rare diseases or conditions meeting the applicable statutory criteria.

Step 4 – Preclinical/Nonclinical Testing

Before clinical use or marketing, applicable devices may require testing such as:

- Mechanical testing
- Electrical safety testing
- Biocompatibility
- Software verification
- Electromagnetic compatibility
- Sterilization validation
- Packaging testing
- Performance testing
- Animal testing where appropriate

The exact testing depends on the device.

Step 5 – Clinical Evaluation/Investigation

For devices requiring clinical evidence, clinical investigations may be conducted to evaluate:

- Safety
- Effectiveness
- Performance
- Patient outcomes
- Adverse events

Where applicable, an **Investigational Device Exemption (IDE)** permits certain investigational devices to be used in clinical studies in accordance with FDA requirements.

Step 6 – Quality Management System

Manufacturers must establish an appropriate quality system.

A major current development is that FDA's **Quality Management System Regulation (QMSR)** became effective on **February 2, 2026**. QMSR amended 21 CFR Part 820 and incorporates by reference **ISO 13485:2016** for medical device quality management systems.

Therefore, students should now understand:

21 CFR Part 820 → QMSR → incorporates ISO 13485:2016

The QMS supports areas such as:

- Design and development
- Manufacturing
- Quality control
- Documentation

- Corrective and preventive actions
- Complaint handling
- Risk management
- Records

Step 7 – Prepare Regulatory Submission

The manufacturer prepares the appropriate regulatory submission.

Depending on the pathway, this may include:

- Device description
- Intended use
- Indications for use
- Design information
- Risk information
- Performance data
- Test results
- Clinical data where required
- Manufacturing information
- Quality-system information
- Labeling

Step 8 – Labeling

Medical device labeling is an important regulatory requirement.

FDA's general device labeling requirements are contained in **21 CFR Part 801**.

Labeling may include:

- Device name
- Manufacturer information
- Intended use
- Indications
- Instructions for use
- Warnings
- Precautions
- Contraindications

- Installation instructions
- Maintenance information

Labeling must not provide misleading information or unsupported claims.

Step 9 – FDA Review

FDA reviews the submitted information according to the applicable pathway.

The review may involve:

- Technical evaluation
- Clinical evaluation
- Statistical evaluation
- Manufacturing/quality considerations
- Labeling review
- Risk assessment

FDA may request additional information where necessary.

Step 10 – Regulatory Decision

Depending on the pathway, the outcome may be:

- 510(k) clearance
- De Novo classification
- PMA approval
- HDE approval
- Exemption where applicable

Step 11 – Post-Market Requirements

FDA oversight does not end when a device reaches the market.

Post-market activities may include:

- Complaint handling
- Medical device reporting
- Corrective actions
- Recalls where necessary
- Post-market surveillance
- Manufacturing quality monitoring

- Monitoring of device safety and performance

FDA states that it is responsible for assuring that medical devices available in the United States remain safe and effective throughout their total product lifecycle.

Simplified FDA Pathway

Medical Device Idea

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Define Intended Use

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Classify Device

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Identify Regulatory Pathway

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Design & Development

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Risk Management

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Nonclinical/Performance Testing

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Clinical Evaluation – where required

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Quality Management System

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Prepare Submission

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FDA Review

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Clearance / Approval / Classification / Exemption

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Manufacturing & Market Launch

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Post-Market Surveillance

Important Difference: Clearance vs Approval

Term	Common pathway	Meaning
FDA Clearance	510(k)	Device has been determined substantially equivalent to an appropriate predicate
FDA Approval	PMA	FDA has approved the PMA based on applicable evidence of safety and effectiveness
De Novo Classification	De Novo	New device type classified through the De Novo pathway
Exempt	Certain devices	No applicable premarket submission, subject to the conditions and limitations of the exemption

5. Reimbursement

Introduction

Reimbursement refers to the payment made to healthcare providers or organizations for eligible healthcare services, procedures, or products.

For a medical device company, reimbursement is important because a product may be clinically effective but commercially unsuccessful if customers cannot obtain adequate financial coverage for its use.

Major Participants

- Patient
- Doctor
- Hospital
- Insurance company/payer
- Government
- Medical device manufacturer

Basic Process

Patient receives treatment

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Healthcare provider documents service

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Claim is submitted

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Payer evaluates claim

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Claim approved/rejected

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Payment made

Importance of Reimbursement

Reimbursement influences:

- Patient affordability
- Hospital purchasing
- Product adoption
- Market size
- Revenue
- Product pricing

Example

A new cardiac monitoring device may provide excellent clinical benefits. However, if hospitals or patients cannot recover the cost through an applicable payment mechanism, adoption may remain low.

6. Managing Issues

During product development and commercialization, several issues can arise.

Common issues

- Technical failures
- Component shortages
- Design changes
- Budget overruns
- Regulatory delays
- Testing failures
- Customer complaints
- Manufacturing problems
- Software errors
- Supplier delays

Issue Management Process

Identify issue → Analyze cause → Determine impact → Prioritize → Develop corrective action → Assign responsibility → Implement action → Monitor → Close issue

Example

If a prototype shows inaccurate SpO_2 readings:

Problem: Inaccurate measurement
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Investigation: Sensor placement/software/calibration
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Root cause: Incorrect calibration algorithm
↓
Action: Modify algorithm
↓
Retest: Verify accuracy
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Closure: Document the corrective action

7. Meeting Deadlines

Meeting deadlines means completing project activities within the planned time.

Methods

1. Develop a project schedule

Define:

- Activities
- Start date
- End date
- Dependencies
- Responsible person

2. Prioritize critical activities

Identify activities that directly affect the final delivery date.

3. Monitor progress

Use:

- Gantt charts
- Milestone tracking
- Weekly meetings
- Progress reports

4. Allocate resources properly

Ensure availability of:

- Personnel
- Equipment
- Materials

- Budget

5. Manage risks

Identify potential delays before they occur.

6. Take corrective action

If an activity is delayed:

- Reallocate resources
- Change sequence
- Add manpower
- Revise schedule
- Address technical problems

Example

If clinical testing is delayed by two weeks, the team should assess whether documentation, software development, or manufacturing activities can proceed in parallel without compromising quality.

8. Product Launch

Introduction

Product launch is the process of introducing a new product into the target market and making it available to customers.

For a medical device, product launch is more than simply selling the product. It involves:

- Regulatory readiness
- Manufacturing readiness
- Market preparation
- Pricing
- Distribution
- Sales training
- Customer support
- Product communication
- Post-market monitoring

Stages of Product Launch

1. Market Preparation

Before launching the product, the company should understand:

- Target customers

- Market size
- Competitors
- Customer requirements
- Pricing
- Distribution channels
- Market barriers

2. Regulatory Readiness

The company must ensure that applicable regulatory requirements have been satisfied before marketing the device in the relevant jurisdiction.

This includes appropriate:

- Regulatory status
- Labeling
- Documentation
- Quality systems
- Manufacturing controls

3. Manufacturing Readiness

The company should ensure that sufficient quantities can be produced with consistent quality.

Activities include:

- Supplier qualification
- Production planning
- Quality inspection
- Packaging
- Sterilization where required
- Inventory planning

4. Pricing Strategy

The company determines the price based on:

- Manufacturing cost
- Competitor prices
- Customer willingness to pay
- Clinical value

- Distribution cost
- Market positioning
- Reimbursement environment

5. Sales Preparation

The sales team should understand:

- Product features
- Clinical benefits
- Product limitations
- Target customers
- Competitors
- Pricing
- Demonstration procedure
- Customer objections

6. Marketing and Promotion

Promotion may include:

- Product demonstrations
- Conferences
- Medical exhibitions
- Digital marketing
- Scientific publications
- Healthcare professional education
- Sales presentations

Medical device claims should remain consistent with applicable regulatory requirements and authorized/intended uses.

7. Distribution

The company determines how the product will reach customers.

Possible channels include:

Manufacturer → Distributor → Hospital → Patient

or

Manufacturer → Hospital

or

Manufacturer → Clinic → Patient

8. Product Training

Healthcare professionals may need training on:

- Installation
- Operation
- Maintenance
- Safety
- Troubleshooting
- Clinical use

9. Launch

The product is officially introduced to the target market.

Launch activities may include:

- Product announcement
- Demonstrations
- Distributor activation
- Sales campaigns
- Hospital visits
- Customer training

10. Post-Launch Monitoring

After launch, the company monitors:

- Sales
- Customer feedback
- Complaints
- Device failures
- Returns
- Adverse events
- Market acceptance
- Profitability

Product Launch Framework

Market Research

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Target Customer Identification

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Regulatory Readiness

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Manufacturing Readiness

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Pricing

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Sales Training

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Marketing & Promotion

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Distribution

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Product Launch

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Customer Support

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Post-Market Monitoring

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Product Improvement

Factors Determining Successful Product Launch

Technical readiness

The device should perform according to its specifications.

Clinical readiness

Healthcare professionals should understand its clinical value.

Regulatory readiness

Required regulatory requirements must be satisfied.

Financial readiness

The company should have adequate financial resources.

Manufacturing readiness

Production should meet expected demand.

Sales readiness

Sales personnel should be properly trained.

Customer support

Installation, training, maintenance, and troubleshooting should be available.

Market readiness

There should be sufficient customer demand.

9. Forecasting

Forecasting is the process of **estimating future demand, sales, revenue, or market requirements**.

It helps companies determine how many products they should manufacture and how much inventory they should maintain.

Types of Forecasting

1. Qualitative forecasting

Based on expert opinion and market knowledge.

Methods include:

- Expert opinion
- Sales-force opinion
- Customer surveys
- Delphi method

2. Quantitative forecasting

Uses historical data and mathematical/statistical techniques.

Examples:

- Moving averages
- Trend analysis
- Regression analysis
- Time-series analysis

Factors Affecting Medical Device Forecasting

- Number of hospitals

- Patient population
- Disease prevalence
- Competitor products
- Product price
- Healthcare policies
- Reimbursement
- Technology adoption
- Economic conditions

Example

If a company sold:

- 500 units in Year 1
- 700 units in Year 2
- 900 units in Year 3

The company may use historical trends and market growth assumptions to estimate future demand.

Importance

Forecasting helps with:

- Production planning
- Inventory control
- Manpower planning
- Financial planning
- Sales targets
- Supply-chain management

10. Pricing

Pricing is the process of determining the **amount a customer must pay for a product or service**.

Objectives of Pricing

- Recover costs
- Generate profit
- Gain market share
- Compete effectively
- Reflect product value
- Encourage adoption

Factors Affecting Pricing

1. Manufacturing cost

Higher production costs generally increase the minimum viable selling price.

2. Research and development cost

Medical device companies must recover development investment over time.

3. Competitor pricing

The company should understand prices of competing products.

4. Customer willingness to pay

Customers may pay more for significant clinical and economic benefits.

5. Clinical value

A product that significantly improves outcomes may justify a higher price.

6. Reimbursement

The available payment mechanism can strongly influence practical affordability and adoption.

7. Distribution cost

Distributor margins, transportation, installation, and service affect the final price.

Common Pricing Approaches

Cost-plus pricing:
Cost + desired profit margin

Competition-based pricing:
Price based on competitors

Value-based pricing:
Price based on perceived/ demonstrated customer and clinical value

Penetration pricing:
Lower initial price to encourage market adoption

Premium pricing:
Higher price for differentiated products with strong value or positioning

11. Product and Sales Support

Introduction

Product and sales support refers to the technical, clinical, marketing, training, and service assistance provided to customers and sales teams before and after a product is sold.

Product Support

Product support includes:

- Installation

- User training
- Operation manuals
- Technical assistance
- Preventive maintenance
- Corrective maintenance
- Spare parts
- Software updates
- Troubleshooting
- Warranty service

Sales Support

Sales support helps sales personnel successfully communicate and demonstrate the product.

It includes:

- Product brochures
- Technical specifications
- Demonstration units
- Product presentations
- Training for sales personnel
- Competitor comparison
- Frequently asked questions
- Clinical evidence
- Customer case studies
- Pricing information

Importance

1. Improves customer confidence

Customers feel more confident when technical support is available.

2. Increases sales effectiveness

Sales personnel can answer technical and clinical questions.

3. Reduces product misuse

Proper training helps users operate devices correctly.

4. Improves customer satisfaction

Fast service and troubleshooting improve the customer experience.

5. Encourages repeat purchases

Satisfied customers are more likely to purchase additional products.

Example

For an ultrasound system, sales support may include product demonstration and clinical application training, while product support includes installation, preventive maintenance, troubleshooting, and service.

12. Managing an Existing Product Line

Introduction

A **product line** is a group of related products marketed by the same company.

For example, a medical technology company may have a product line containing:

- Basic patient monitor
- Advanced patient monitor
- ICU monitor
- Wireless monitor

Managing an existing product line means continuously monitoring and improving the products to maintain their **market position, profitability, quality, and customer satisfaction**.

Major Activities

1. Monitor sales performance

Track:

- Sales volume
- Revenue
- Profit
- Market share
- Growth rate

2. Monitor customer feedback

Collect information about:

- Product satisfaction
- Complaints
- Usability
- Reliability
- Desired improvements

3. Product improvement

Improve:

- Hardware
- Software
- User interface
- Accuracy
- Connectivity
- Reliability

4. Manage product variants

Companies may offer different versions for different customer segments.

5. Manage pricing

Prices may need adjustment based on:

- Competition
- Cost
- Demand
- Market position

6. Inventory management

Maintain sufficient stock without creating excessive inventory.

7. Competitor monitoring

Track:

- New competitor products
- Technology changes
- Pricing
- Features
- Market strategy

8. Product life-cycle management

A product generally passes through:

Introduction → Growth → Maturity → Decline

At each stage, the company may modify its marketing and product strategy.

9. Decide whether to modify or discontinue

If a product has declining demand, high maintenance costs, or is replaced by better technology, the company may:

- Improve it
- Reposition it
- Reduce its price

- Replace it
- Discontinue it